Stream: Internet Engineering Task Force (IETF)
RFC: 10079
Updates: 9605
Category: Standards Track
Published: November 2026
ISSN: 2070-1721
Authors: R. Barnes, *Cisco*; E. Omara, *Apple*; A. Rosenberg, *Apple*

RFC 10079
Updates to the SFrame Cipher Suites Registry

Abstract

This document addresses two omissions in the Secure Frame (SFrame) protocol specification. First, the definition of the IANA "SFrame Cipher Suites" registry in RFC 9605 omits several important fields. This document updates the IANA registry by adding those fields and defining the contents of those fields for current entries. Second, the Authenticated Encryption with Associated Data (AEAD) construction based on AES-CTR and HMAC is defined only for the 128-bit security level. This document registers parallel constructions at the 256-bit security level.

Status of This Memo

This is an Internet Standards Track document.

This document is a product of the Internet Engineering Task Force (IETF). It represents the consensus of the IETF community. It has received public review and has been approved for publication by the Internet Engineering Steering Group (IESG). Further information on Internet Standards is available in Section 2 of RFC 7841.

Information about the current status of this document, any errata, and how to provide feedback on it may be obtained at https://www.rfc-editor.org/info/rfc10079.

Copyright Notice

Copyright (c) 2026 IETF Trust and the persons identified as the document authors. All rights reserved.

This document is subject to BCP 78 and the IETF Trust's Legal Provisions Relating to IETF Documents (https://trustee.ietf.org/license-info) in effect on the date of publication of this document. Please review these documents carefully, as they describe your rights and restrictions with respect to this document. Code Components extracted from this document must include Revised BSD License text as described in Section 4.e of the Trust Legal Provisions and are provided without warranty as described in the Revised BSD License.

Table of Contents

1. Introduction 2
2. AES-256-CTR with HMAC-SHA512 2
3. Security Considerations 3
4. IANA Considerations 3
4.1. Updates to the SFrame Cipher Suites Registry 3
4.2. Cipher Suites for AES-256-CTR with HMAC-SHA512 4
5. References 5
5.1. Normative References 5
5.2. Informative References 5
Appendix A. Test Vectors 5
A.1. AEAD Encryption/Decryption Using AES-CTR and HMAC 5
A.2. SFrame Encryption/Decryption 6
Authors' Addresses 9

1. Introduction

This document addresses two omissions in the Secure Frame (SFrame) protocol specification [RFC9605]. First, in Section 8.1 of [RFC9605], the definition of the IANA "SFrame Cipher Suites" registry omits several important fields. This document defines the required fields and the contents of those fields for current entries. IANA has updated the registry accordingly. Second, the AEAD construction based on AES-CTR and HMAC (in Section 4.5.1 of [RFC9605]) is defined only for the 128-bit security level. This document registers parallel constructions at the 256-bit security level.

2. AES-256-CTR with HMAC-SHA512

Section 4.5.1 of [RFC9605] defines a compound authenticated encryption construction, using the unauthenticated Counter (CTR) mode of AES for encryption and HMAC for authentication.

The original specification only defines cipher suite values for instances of this construction using AES-128-CTR and HMAC-SHA256. The construction works the same way when used with AES-256-CTR and HMAC-SHA512. The only differences are in the lengths of some SFrame intermediate values:

- The keys generated by SFrame key derivation (`derive_key_salt`) are longer to match the needs of AES-256-CTR and HMAC-SHA512 (96 bytes vs. 48 bytes for AES-128-CTR and HMAC-SHA256).
- The initial tag value `tag` in `compute_tag` is 64 bytes instead of 32 bytes.

Identifiers for cipher suites using AES-256-CTR and HMAC-SHA512 are defined in Section 4.2.

3. Security Considerations

The registry changes in this document have no effect on the security of SFrame.

The new algorithms registered by this document allow the construction combining AES-CTR with HMAC to be used in environments that require a 256-bit security level.

4. IANA Considerations

Per this document, IANA has updated the columns in the "SFrame Cipher Suites" registry (see Section 4.1), added entries to the updated registry for the new cipher suites defined in this document (see Section 4.2), and added this document as an additional reference for this registry.

4.1. Updates to the SFrame Cipher Suites Registry

IANA has added the following columns to the "SFrame Cipher Suites" registry. This document also updates the registration template in Section 8.1 of [RFC9605] to include these fields, which are placed between the "Name" and "Recommended" fields.

- `Nh`: The size in bytes of the output of the hash function.
- `Nka`: For cipher suites using the compound AEAD described in Section 4.5.1 of [RFC9605], the size in bytes of a key for the underlying encryption algorithm; for other cipher suites, the value is "n/a".
- `Nk`: The size in bytes of a key for the encryption algorithm.
- `Nn`: The size in bytes of a nonce for the encryption algorithm.
- `Nt`: The overhead in bytes of the encryption algorithm (typically the size of a "tag" that is added to the plaintext).

The registration procedures in Section 8.1 of [RFC9605] are otherwise unchanged.

Table 1 illustrates the new structure of the registry and provides the required values for the previously registered entries.

Value	Name	Nh	Nka	Nk	Nn	Nt	R	Reference	Ch Contr
0x0000	Reserved	-	-	-	-	-	-	RFC 9605	IETF
0x0001	AES_128_CTR_HMAC_SHA256_80	32	16	48	12	10	Y	RFC 9605, RFC 10079	IETF
0x0002	AES_128_CTR_HMAC_SHA256_64	32	16	48	12	8	Y	RFC 9605, RFC 10079	IETF
0x0003	AES_128_CTR_HMAC_SHA256_32	32	16	48	12	4	Y	RFC 9605, RFC 10079	IETF
0x0004	AES_128_GCM_SHA256_128	32	n/a	16	12	16	Y	RFC 9605, RFC 10079	IETF
0x0005	AES_256_GCM_SHA512_128	64	n/a	32	12	16	Y	RFC 9605, RFC 10079	IETF
0xF000 - 0xFFFF	Reserved for Private Use	-	-	-	-	-	-	RFC 9605	IETF

Table 1: New Structure and Contents of the SFrame Cipher Suites Registry

4.2. Cipher Suites for AES-256-CTR with HMAC-SHA512

IANA has added the following new entries to the "SFrame Cipher Suites" registry:

Value	Name	Nh	Nka	Nk	Nn	Nt	R	Reference	Ch Contr
0x0006	AES_256_CTR_HMAC_SHA512_80	64	32	96	12	10	Y	RFC 10079	IETF
0x0007	AES_256_CTR_HMAC_SHA512_64	64	32	96	12	8	Y	RFC 10079	IETF
0x0008	AES_256_CTR_HMAC_SHA512_32	64	32	96	12	4	Y	RFC 10079	IETF

Table 2: New Entries in the SFrame Cipher Suites Registry

5. References

5.1. Normative References

[RFC9605] Omara, E., Uberti, J., Murillo, S. G., Barnes, R., Ed., and Y. Fablet, "Secure Frame (SFrame): Lightweight Authenticated Encryption for Real-Time Media", RFC 9605, DOI 10.17487/RFC9605, August 2024, <https://www.rfc-editor.org/info/rfc9605>.

5.2. Informative References

[TestVectors] "test-vectors-aes256.json", commit 78680b9, 18 September 2025, <https://github.com/bifurcation/sframe-iana-256/blob/main/test-vectors/test-vectors-aes256.json>.

Appendix A. Test Vectors

This appendix provides a set of test vectors that implementations can use to verify that they correctly implement SFrame encryption and decryption with the cipher suites registered in this document. Test vectors are provided for both the AES-256-CTR-HMAC construction and full SFrame encryption with the new cipher suites.

All values are either numeric or byte strings. Numeric values are represented as hex values, prefixed with `0x`. Byte strings are represented in hex encoding.

Line breaks and whitespace within values are inserted to conform to the width requirements of the RFC format. They should be removed before use.

These test vectors are also available in JSON format at [TestVectors]. In the JSON test vectors, numeric values are JSON numbers and byte string values are JSON strings containing the hex encoding of the byte strings.

A.1. AEAD Encryption/Decryption Using AES-CTR and HMAC

For each case, we provide:

- `cipher_suite`: The index of the cipher suite in use (see Section 4.1)
- `key`: The `key` input to encryption/decryption
- `enc_key`: The encryption subkey produced by the `derive_subkeys()` algorithm
- `auth_key`: The authentication subkey produced by the `derive_subkeys()` algorithm
- `nonce`: The `nonce` input to encryption/decryption
- `aad`: The `aad` input to encryption/decryption
- `pt`: The plaintext
- `ct`: The ciphertext

An implementation should verify that the following are true, where `AEAD.Encrypt` and `AEAD.Decrypt` are as defined in Section 4.5.1 of [RFC9605]:

- `AEAD.Encrypt(key, nonce, aad, pt) == ct`
- `AEAD.Decrypt(key, nonce, aad, ct) == pt`

The other values in the test vector are intermediate values provided to facilitate debugging of test failures.

A.2. SFrame Encryption/Decryption

For each case, we provide:

- `cipher_suite`: The index of the cipher suite in use (see Section 4.1)
- `kid`: A Key ID (KID) value
- `ctr`: A Counter (CTR) value
- `base_key`: The `base_key` input to the `derive_key_salt` algorithm
- `sframe_key_label`: The label used to derive `sframe_key` in the `derive_key_salt` algorithm
- `sframe_salt_label`: The label used to derive `sframe_salt` in the `derive_key_salt` algorithm
- `sframe_secret`: The `sframe_secret` variable in the `derive_key_salt` algorithm
- `sframe_key`: The `sframe_key` value produced by the `derive_key_salt` algorithm
- `sframe_salt`: The `sframe_salt` value produced by the `derive_key_salt` algorithm
- `metadata`: The `metadata` input to the SFrame `encrypt` algorithm
- `pt`: The plaintext
- `ct`: The SFrame ciphertext

An implementation should verify that the following are true, where `encrypt` and `decrypt` are as defined in Section 4.4 of [RFC9605], using an SFrame context initialized with `base_key` assigned to `kid`:

- `encrypt(ctr, kid, metadata, plaintext) == ct`
- `decrypt(metadata, ct) == pt`

The other values in the test vector are intermediate values provided to facilitate debugging of test failures.

```
cipher_suite: 0x0006
kid: 0x0000000000000123
ctr: 0x0000000000004567
base_key: 000102030405060708090a0b0c0d0e0f
sframe_key_label: 534672616d6520312e30205365637265
                  74206b6579200000000000000001230006
sframe_salt_label: 534672616d6520312e30205365637265
                   742073616c7420000000000000012300
                   06
sframe_secret: 0fc3ea6de6aac97a35f194cf9bed94d4
               b5230f1cb45a785c9fe5dce9c188938a
               b6ba005bc4c0a19181599e9d1bcf7b74
               aca48b60bf5e254e546d809313e083a3
sframe_key: 3c343886ec1c79278836863e00fe934c
            8894460cfa367ebdc4856b0a9268a4f4
            fb99437876819394ef90b10ee12602d0
            23f7128ee50f2314c2cc3cff4c56616d
            2fe03ad2a254cc2ed29b2a4d3f2534c0
            dda9e7c391ad1917ea07aa221dd4b224
sframe_salt: e082f7ce012ad30c87c49e3f
metadata: 4945544620534672616d65205747
nonce: e082f7ce012ad30c87c4db58
aad: 990123456749455446205346726 16d65
     205747
pt: 64726166742d696574662d736672616d
    652d656e63
ct: 9901234567b369e03ec6467ad505ddc8
    4914115069280c5c797555be6e32cde6
    ac25bc9e
```

```
cipher_suite: 0x0007
kid: 0x0000000000000123
ctr: 0x0000000000004567
base_key: 000102030405060708090a0b0c0d0e0f
sframe_key_label: 534672616d6520312e30205365637265
                  74206b6579200000000000000001230007
sframe_salt_label: 534672616d6520312e30205365637265
                   742073616c7420000000000000012300
                   07
sframe_secret: 0fc3ea6de6aac97a35f194cf9bed94d4
               b5230f1cb45a785c9fe5dce9c188938a
               b6ba005bc4c0a19181599e9d1bcf7b74
               aca48b60bf5e254e546d809313e083a3
sframe_key: 7271d6c6cbccd2e2343d480ebea65718
            a7bb379eefcf3f8d107c1e2a76e75529
            3a497fd9e4e8291b965161987ef4ef24
            983eabb06cb0a392defaab18654780a3
            9c106ffa4a47d4183a6e593cd0c1bcab
            2b9c6dcf049215845bfb7580c4dea80e
sframe_salt: 46b4367993a314910d4d9f3d
metadata: 4945544620534672616d65205747
nonce: 46b4367993a314910d4dda5a
aad: 990123456749455446205346726 16d65
     205747
pt: 64726166742d696574662d736672616d
    652d656e63
ct: 990123456797cb5644d8831ff8bdc080
    249990b24b569144cab2a87be22c20d9
    7976
```

```
cipher_suite: 0x0008
kid: 0x0000000000000123
ctr: 0x0000000000004567
base_key: 000102030405060708090a0b0c0d0e0f
sframe_key_label: 534672616d6520312e30205365637265
                  74206b6579200000000000000001230008
sframe_salt_label: 534672616d6520312e30205365637265
                   742073616c74200000000000000012300
                   08
sframe_secret: 0fc3ea6de6aac97a35f194cf9bed94d4
               b5230f1cb45a785c9fe5dce9c188938a
               b6ba005bc4c0a19181599e9d1bcf7b74
               aca48b60bf5e254e546d809313e083a3
sframe_key: afe92c81e0df8c00fab619e0559fe5ae
            efce1ef77789d4c728af1b1c1f2e3552
            c405d274415a5291ec075c2d9954c450
            fbd36682a4e978494808b703ce78b409
            f9fec29b91e6e703a75c4131377c80c9
            d51b8906088092452e2593eb142eea2d
sframe_salt: f6de647bac1263524cfb6533
metadata: 4945544620534672616d65205747
nonce: f6de647bac1263524cfb2054
aad: 990123456749455446205346726 16d65
     205747
pt: 64726166742d696574662d736672616d
    652d656e63
ct: 9901234567112a94a288b85b49ffef1d
    279f2830165c39d76cac8884011c
```

Authors' Addresses

Richard Barnes
Cisco
Email: rlb@ipv.sx

Emad Omara
Apple
Email: emad.omara@gmail.com

Aron Rosenberg
Apple
Email: aron.rosenberg@apple.com